

Thevenin

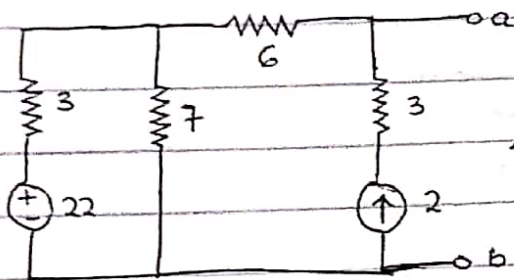
* R_{th}

* V_{th}

> $R_{th} = R_N = R_L$

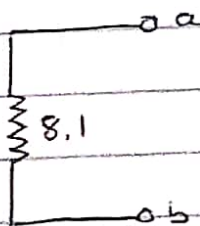
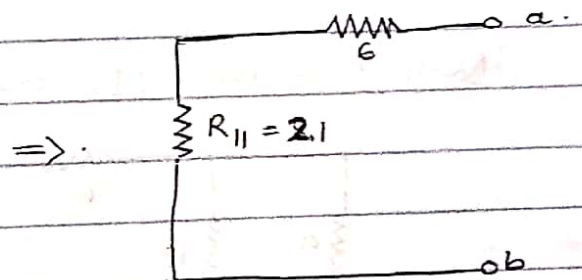
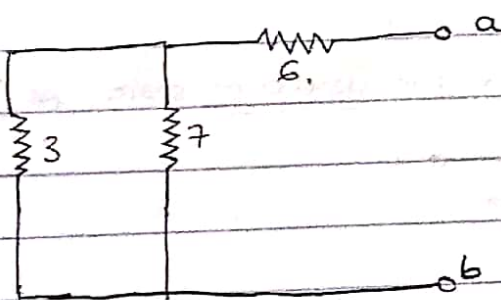
> $R_{\text{thevenin}} = R_{\text{Norton}}$

R-thevenin



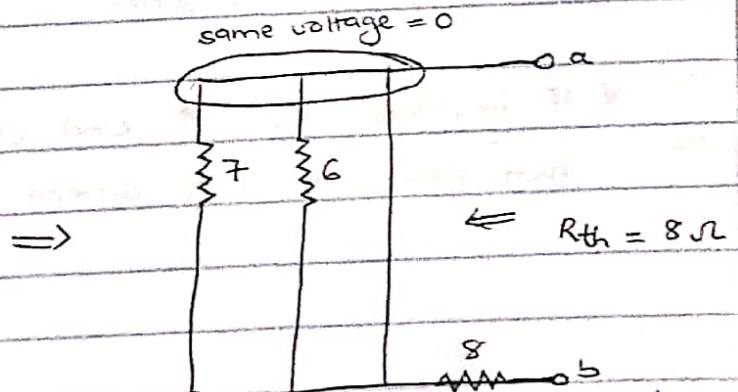
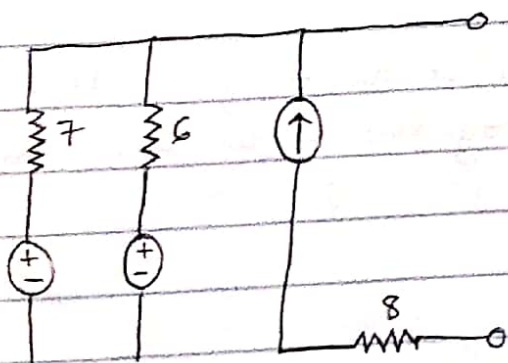
* we will be calculating
R equivalent

* If we are told to calculate R-thevenin close a voltage source with a line



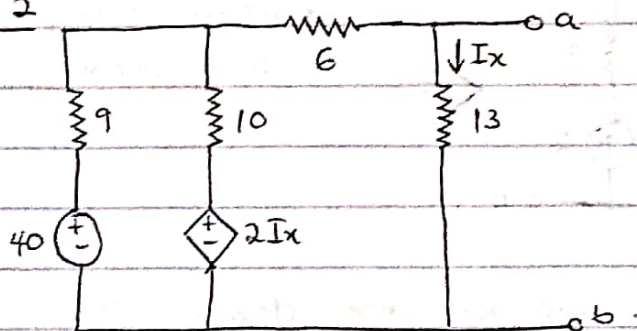
$\therefore R_{th} = 8.1 \text{ ohms}$

Example

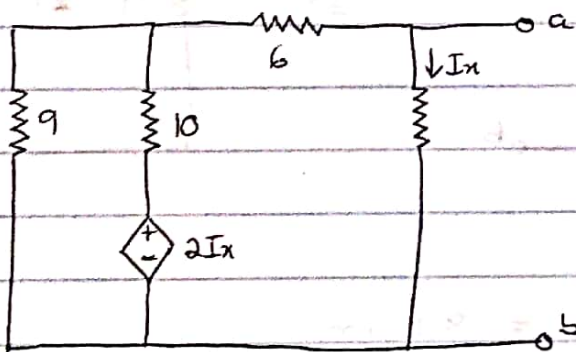


* in parallel with a line hence remove 7Ω & 6Ω
because they are sharing a node with a line

Example 2



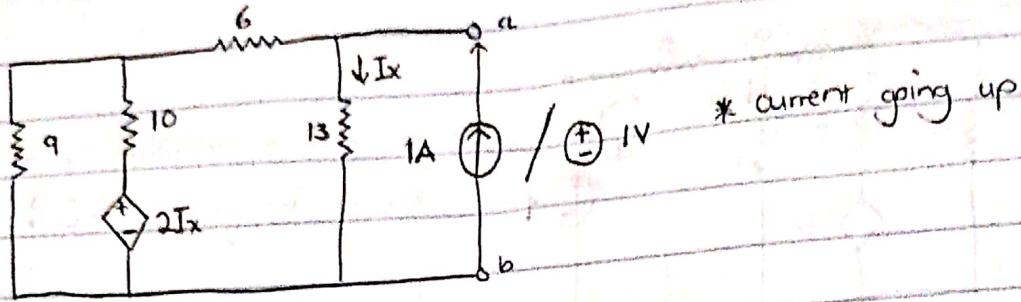
Don't touch the $2I_x$ source i.e. the dependent source. (current / voltage source)



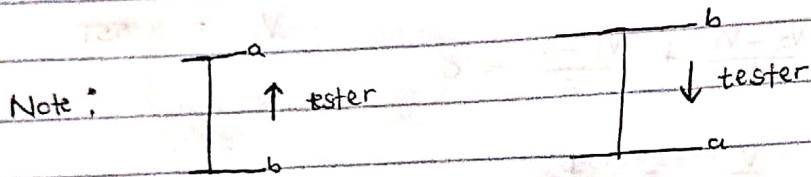
* Once you see a dependent source on your circuit, resistors will not be in parallel or series

* If you have that ↑ kind of circuit we are going to add a TESTER, that tester will be a current or voltage and it will have a magnitude of 1A

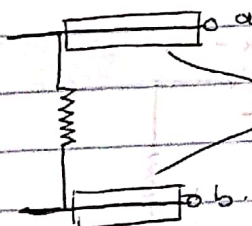
Example



NB: The polarity of the dependent source is not related to the polarity of the TESTER

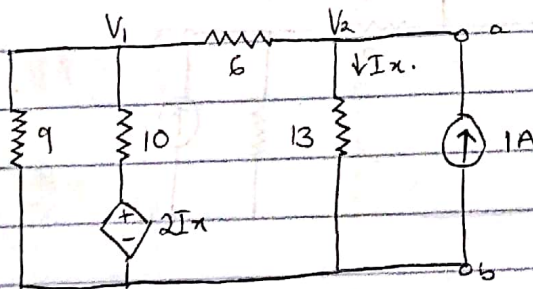


* we are always going to apply $\oplus 1A$ if and only if we don't have a branch between a and b



if we don't have a branch on the indicated (by a box) position and also between a and b

* we only plug $\oplus 1V$ if and only if we have ^{one or more} resistors ~~branch~~ on the boxed positions above.



we have current, therefore we need voltage to calculate R_{th}

$\Rightarrow$ we use nodal analysis.

$$@V_1: \frac{V_1 - V_2}{6} + \frac{V_1}{9} + \frac{V_1 - 2I_x}{10} = 0$$

$$@V_2: \frac{V_2 - V_1}{6} + \frac{V_2}{13} - 1 = 0$$

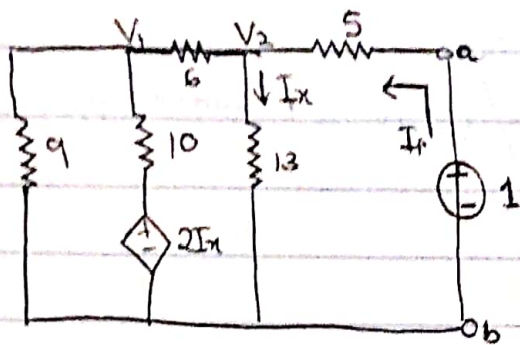
$$\text{but } I_x = \frac{V_2}{13}$$

$$\frac{V_1}{6} - \frac{V_2}{6} + \frac{V_1}{9} + \frac{V_1}{10} - \frac{1}{65} V_2 = 0$$

$$V_1 = 2.95 V$$

$$V_2 = 6.12 V$$

$$\therefore R_{th} = \frac{V_2}{1} = \frac{6.12}{1} = 6.12 \Omega$$



* you need to calculate I_p to make it easier for you. SSC !!

$$I_p = - \left(\frac{V_2 - 1}{5} \right)$$

$$@ V_1 : \frac{V_1}{9} + \frac{V_1 - V_2}{6} + \frac{V_1 - 2 \left(\frac{V_2}{13} \right)}{10} = 0$$

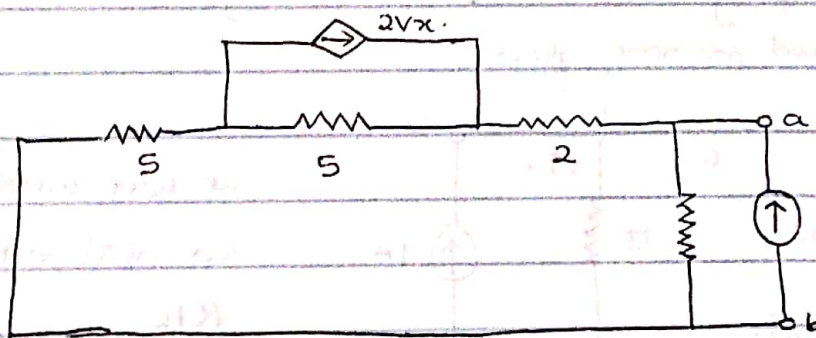
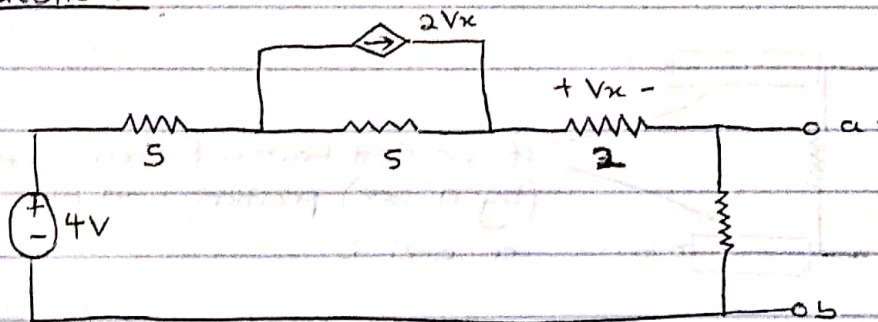
$$V_1 = 0.265$$

$$V_2 = 0.551$$

$$@ V_2 : \frac{V_2}{13} + \frac{V_2 - V_1}{6} + \frac{V_2 - 1}{5} = 0$$

$$\therefore R_{th} = \frac{V}{I_p} = \frac{1}{I_p}$$

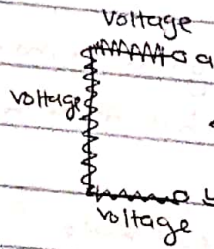
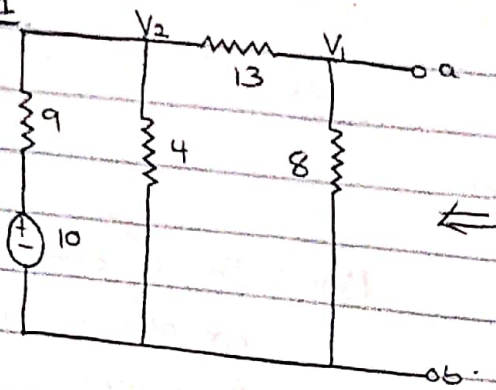
Asked question.



* can do source transformation. (continue)

V_{th} (V-thevenin)

CASE 1



← voltage of the indicated position gives voltage of the gap.

* with V-thevenin (V_{th}) you do not remove any source just do nodal analysis. SSC !!

$$\text{@ } V_1: \frac{V_1 - V_2}{13} + \frac{V_1}{8} = 0$$

$$\frac{V_1}{13} + \frac{V_1}{8} - \frac{V_2}{13} = 0$$

$$\text{@ } V_2: \frac{V_2 - V_1}{13} + \frac{V_2 - 10}{9} + \frac{V_2}{4} = 0$$

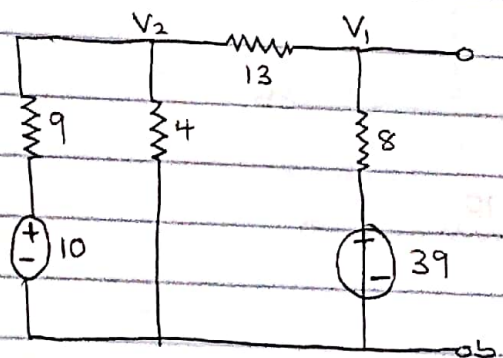
$$\frac{V_2}{13} + \frac{V_2}{9} + \frac{V_2}{4} - \frac{V_1}{13} - \frac{10}{9} = 0$$

$$V_1 = 1.04 \text{ V}$$

$$V_2 = 2.72 \text{ V}$$

$$V_{th} = V_{ab} = V_1$$

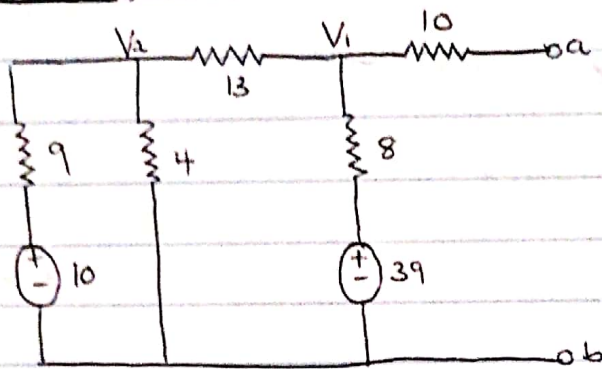
CASE 2



$$\text{@ } V_1: \frac{V_1 - V_2}{13} + \frac{V_1 - 39}{8} = 0$$

then continue with nodal analysis

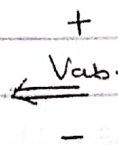
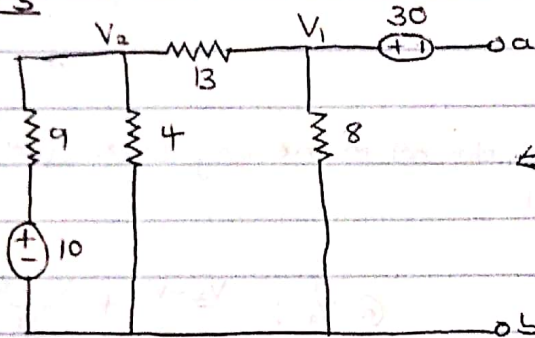
CASE 3



Note : The 10 ohm resistor does not affect anything since its current is 0

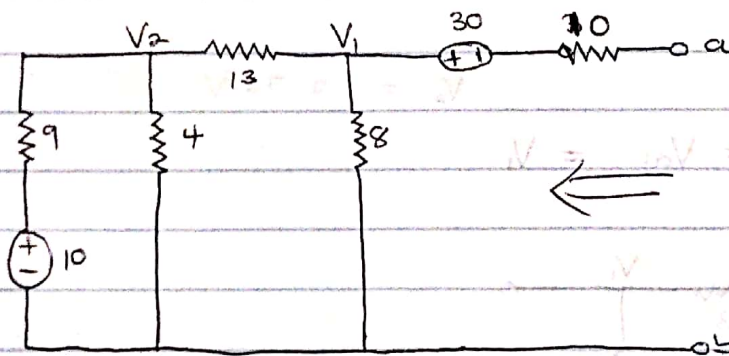
∴ you can remove it or leave it. but its of no use. SSC!!

CASE 5



* supernode.

$$V_{ab} - V_1 = 30$$

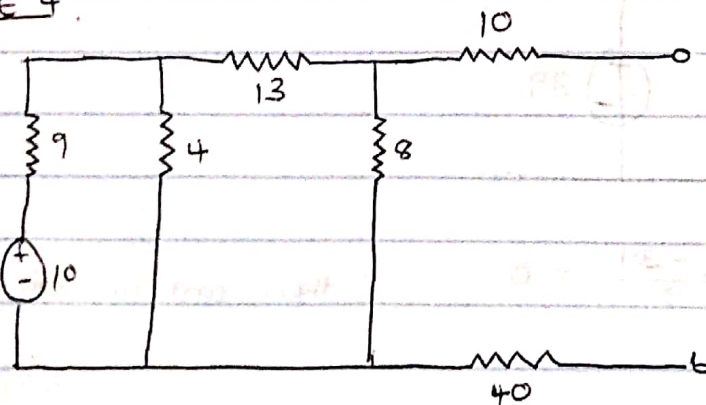


$$\frac{V_1 - 30}{10}$$

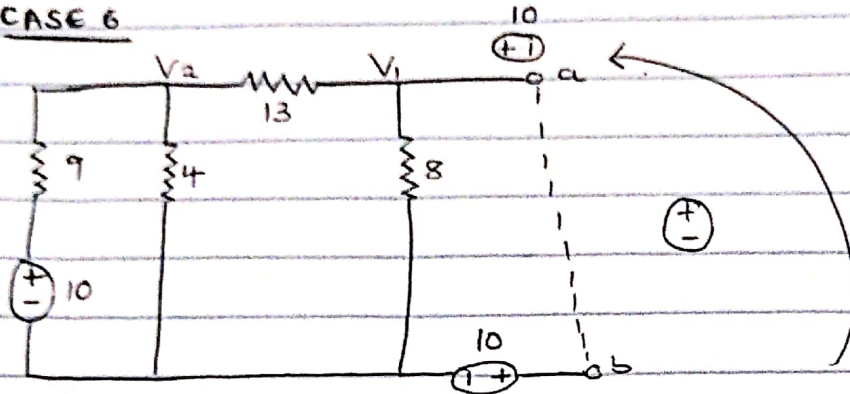
wrong

* there is no current on the gap.

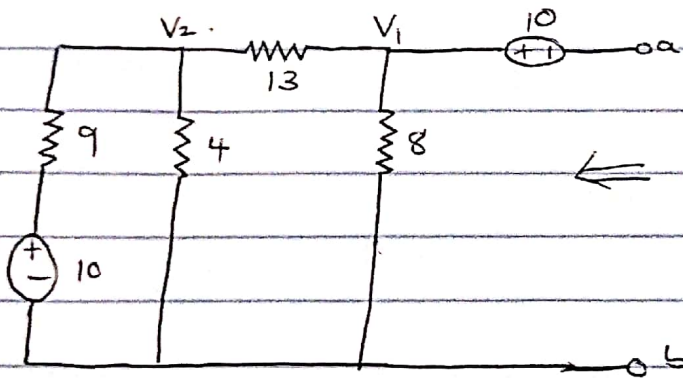
CASE 4



CASE 6



push the 10V source to the top part



$$V_{ab} - V_1 = 10$$

Some complicated examples

